

A Conjecture on Bounded Prime Generation by Polynomials

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Conjecture Statement

Conjecture 2.2:

For every pair of positive integers (k, N) where $k \leq N$, there exists a polynomial $P(n)$ with integer coefficients and degree strictly less than $N-1$ such that $P(n)$ generates **exactly k prime values** when evaluated at $n \in \{0, 1, 2, \dots, N\}$.

Formal Statement

$\forall k, N \in \mathbb{N}^+$ with $k \leq N$, $\exists$ polynomial $P(n) \in \mathbb{Z}[n]$ with $\deg(P) < N-1$ such that:

$$|\{n \in \{0, 1, \dots, N\} : P(n) \text{ is prime}\}| = k$$

Context and Related Work

This conjecture is related to but distinct from several classical results and open problems:

Known Results:

- Dirichlet's Theorem** (1837): Linear polynomials $P(n) = an + b$ with $\gcd(a, b) = 1$ generate infinitely many primes.
- Green-Tao Theorem** (2004): The primes contain arbitrarily long arithmetic progressions, which can be used to construct polynomials generating k consecutive primes via Lagrange interpolation.

Open Conjectures:

- Bunyakovsky Conjecture** (1857): An irreducible polynomial with integer coefficients and $\gcd$ of all coefficients equal to 1 should generate infinitely many primes, provided no fixed prime divides all values.
- Consecutive Prime Generation**: For every k , there exists a polynomial generating k consecutive prime values (essentially proven via Green-Tao interpolation method).

Distinction from Related Problems

This conjecture differs from existing work in a crucial way:

- Classical approach**: Focus on polynomials generating infinitely many or consecutively many primes

- **Proposed approach:** Focus on constructing polynomials to hit an **exact finite target** k within an **arbitrary bound** N

The key difficulty is that k and N are **independent parameters**. Unlike the consecutive prime generation result (where the bound N is determined by k), here we must construct polynomials that:

1. Generate exactly k primes within $[0, N]$
2. Generate only composite values at all other points in $[0, N]$

This requires active control over both prime generation **and** composite generation.

Significance

If true, this conjecture would demonstrate:

- Polynomials can be "tuned" to generate precise numbers of primes in bounded ranges
- There is sufficient flexibility in polynomial construction to avoid unwanted primes
- A deep connection between polynomial degree, coefficients, and prime distribution

If false, it would suggest fundamental limitations on our ability to control prime generation via polynomials, even in finite settings.

Open Questions

1. Is the degree bound $\deg(P) < N-1$ necessary, or could a tighter bound be established?
 2. Can the coefficients of $P(n)$ be bounded polynomially in k and N ?
 3. Is there an algorithm to construct such polynomials?
 4. What is the relationship between k and N that makes construction possible/impossible?
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Status

OPEN - As of January 2026, this conjecture appears to be original and has not been addressed in the mathematical literature.